Physically Induced Atmospheric Adversarial Perturbations: Enhancing Transferability and Robustness in Remote Sensing Image Classification

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***Abstract*—Adversarial attacks pose a severe threat to the reliability of deep learning models in remote sensing (RS) image classification. Most existing methods rely on direct pixelwise perturbations, failing to exploit the inherent atmospheric characteristics of RS imagery or survive real-world image degradations. In this article, we propose FogFool, an atmospherically inspired digital-domain adversarial framework that generates visually plausible fog-based perturbations by iteratively optimizing Perlin noise-initialized atmospheric patterns. By modeling fog formations with natural, irregular structures, FogFool generates adversarial examples that are not only visually consistent with authentic RS scenes but also deceptive. By leveraging the spatial coherence and mid-to-low-frequency nature of atmospheric phenomena, FogFool embeds adversarial information into structural features shared across diverse architectures. Extensive experiments on two benchmark RS datasets demonstrate that FogFool achieves superior performance: not only does it exceed in white-box settings but also exhibits exceptional black-box transferability reaching 83.74% transfer attack success rates (TASRs) and robustness against common preprocessing-based defenses such as JPEG compression (JC) and filtering. Detailed analyses, including confusion matrices and class activation map (CAM) visualizations, reveal that our atmospheric-driven perturbations induce a universal shift in model attention. These results indicate that FogFool represents a practical, stealthy, and highly persistent threat to RS classification systems, providing a robust benchmark for evaluating model reliability in complex environments.**

***Index Terms*—Adversarial attack, adversarial examples, remote sensing (RS) image, scene classification.**

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I. Introduction

With the rapid development of satellite and airborne sensors, vast amounts of remote sensing (RS) imagery with high spatial, spectral, and temporal resolutions are continuously collected [1], providing critical information for a wide range of applications, such as land cover mapping [2], urban planning [3], environmental monitoring [4], and disaster detection [5]. Effectively extracting semantic information from these data is a key challenge in RS image analysis. Among downstream tasks, RS image classification is fundamental, aiming to assign semantic labels to image pixels or regions based on visual and spectral information [6]. Recent advances in deep learning, particularly convolutional neural networks (CNNs), have substantially improved classification performance through automatic feature learning [7]. While these methods achieve impressive results on benchmark datasets [8], their vulnerability to adversarial perturbations raises concerns about robustness and reliability in real-world applications [9], [10].

Adversarial attacks have recently attracted increasing attention in the RS community as deep learning techniques become widely adopted in this field. Czaja et al. [11] conducted one of the earliest empirical studies on adversarial examples in satellite imagery, highlighting the unique challenges of RS scenarios, including viewing geometry, atmospheric effects, and temporal variability. Subsequent studies demonstrated

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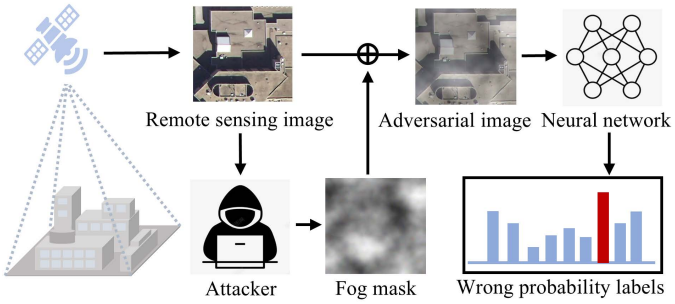


Fig. 1. Overview of the proposed fog-based adversarial attack scenario in RS image classification. An attacker generates a physically plausible fog mask using Perlin noise, which is blended with the original RS image to form an adversarial example. When fed into a deep neural network, this adversarial example causes the model to produce incorrect classification results, posing a security threat to real-world RS systems.

that RS image classification models are highly susceptible to adversarial perturbations under both white-box and black-box settings, posing serious security risks in practical applications. However, most existing attack methods rely on direct pixelwise perturbations and fail to exploit the inherent characteristics of RS imagery, such as ubiquitous atmospheric phenomena. This reliance on high-frequency digital noise often results in limited cross-model transferability and low resistance to image-processing defenses, making them less effective in practical RS pipelines.

Although several studies have explored weather-based attacks, notable limitations persist. For example, Tang et al. [12] simulated snow, fog, shadows, and sun flares to attack optical aerial object detectors; however, their perturbations mainly imitate weather-related color patterns and fail to capture the irregular and natural structures of real clouds and shadows, resulting in visually artificial shapes, e.g., circles or triangles. Sun et al. [13] proposed an adversarial cloud attack for RS salient object detection by constraining images with exposure matrices and additive perturbations, but the generated cloud patterns still exhibit noticeable unnatural noise. Moreover, both methods are restricted to object detection tasks. For RS image classification, Ma et al. [14] introduced a cloud-based adversarial example generation method that uses a Perlin gradient generator network optimized via differential evolution to enable query-efficient black-box attacks. Nevertheless, this approach does not support targeted attacks, lacks controllability over cloud density, and requires additional network training for parameter generation.

Motivated by these limitations, this article investigates adversarial attacks that exploit atmospheric phenomena in RS imagery. As illustrated in Fig. 1, we investigate a digital-domain adversarial threat model in which perturbations are constrained to atmospherically inspired fog-like structures. In this scenario, an attacker optimizes a natural-looking fog mask based on procedural Perlin noise, which is then blended with the original image to create an adversarial example. By modeling fog formations with natural, irregular structures, the generated adversarial examples remain visually consistent with authentic RS images. By shifting the perturbation from isolated pixels to spatially coherent atmospheric structures, FogFool targets the mid-to-low frequency features shared across diverse deep learning architectures, hence significantly enhancing black-box transferability.

Compared to existing methods, the main contributions of this article are summarized as follows.

1) We propose *FogFool*, a fog-based adversarial attack framework for RS image classification that generates targeted and untargeted adversarial examples by optimizing fog patterns using Perlin noise, thereby producing structured, visually natural perturbations.
2) We design a controllable fog-density parameterization that enables fine-grained control over perturbation strength and facilitates the analysis of the tradeoff between visual realism and attack effectiveness.
3) Extensive experiments on two benchmark RS datasets validate that FogFool yields superior attack performance with 97.67%/99.93% targeted success rates on UC Merced land use (UCM)/NWPU and 83.74% black-box transfer rate on NWPU while exhibiting stronger robustness against JPEG compression (JC) and filtering defenses by embedding adversarial information into global atmospheric distributions.

The remainder of this article is organized as follows. Section II reviews related work on adversarial attacks. Section III details the proposed FogFool framework. Section IV presents experimental settings and results. Section V concludes this article.

II. Related Work

A. Adversarial Attacks on Deep Neural Networks

Adversarial examples were first revealed by [15], which demonstrated that imperceptible perturbations can cause deep neural networks to make incorrect predictions with high confidence. Since then, extensive studies have explored the vulnerability of deep models to adversarial examples. Existing attack methods are generally categorized into optimization-based approaches, such as L-BFGS [15] and C&W attacks [16], and gradient-based approaches, including FGSM [17], its iterative variants such as BIM [18] and PGD [19], momentum-based methods [20], and sparse attacks [21]. Beyond digital-domain attacks, recent studies have explored physically realizable and visually natural adversarial perturbations, e.g., adversarial stickers [22], shadow-based attacks [23], weather-induced disturbances, including rain and snow [24], and adversarial camouflage that blends perturbations into natural textures and styles [25], [26].

B. Adversarial Attacks in RS

Under the white-box assumption, several studies have systematically investigated the vulnerability of RS image classification models. Chen et al. [27] first evaluated classical gradient-based attacks on CNN-based RS image recognition models, revealing their sensitivity to adversarial perturbations. They further observed that adversarial misclassifications tend to be selective, with predictions biased toward semantically

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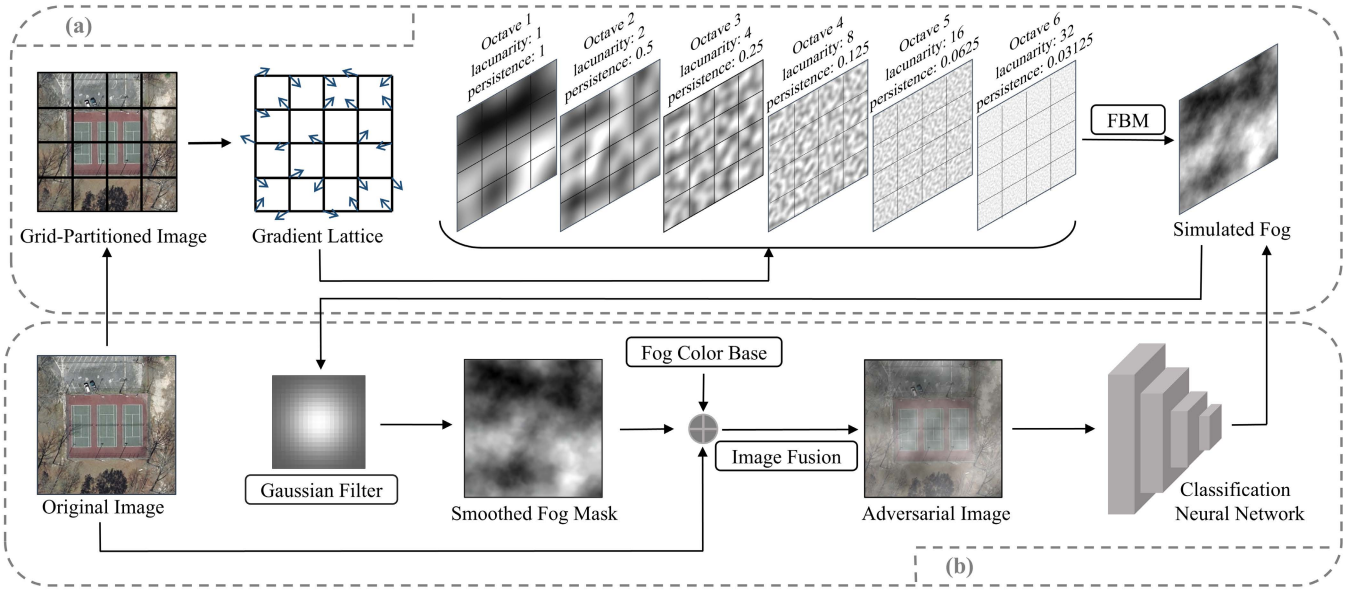


Fig. 2. Overview of the proposed method for fog-based adversarial example generation, including (a) procedural fog simulation module and (b) gradient-guided fog adversarial optimization module. The first module initializes a structured fog mask from multioctave Perlin noise, while the second module iteratively optimizes the fog mask using model gradients to generate the final adversarial example.

similar classes. Subsequent large-scale empirical analyses confirmed the prevalence of adversarial examples across different network architectures and datasets [28], and demonstrated their strong transferability across models [29].

In contrast to white-box attacks, black-box attacks are more practical in real-world scenarios, where the internal structure and parameters of the target model are inaccessible. A key strategy for black-box adversarial attacks is to enhance the transferability of adversarial examples across different models [30]. Xu and Ghamisi [31] pioneered the study of universal adversarial examples in RS, proposing black-box attack methods that exploit shared vulnerabilities among models and establishing a benchmark for evaluation. Building upon this line of research, Bai et al. [32] introduced targeted variants of universal adversarial examples, enabling more fine-grained and controllable attacks. More recently, Wang et al. [33] proposed a black-box attack framework based on feature approximation that improves perturbation precision and cross-model transferability. These studies have collectively demonstrated that black-box adversarial attacks pose a serious and realistic threat to RS image classification models [34]. In parallel, privacy-preserving mechanisms [35], [36], [37] have been explored to address robustness and accountability in machine learning systems.

III. Atmospherically Inspired Adversarial Perturbation Design

A. Overview of the Proposed Method

As illustrated in Fig. 2, we propose a fog-based adversarial attack framework, namely FogFool, that exploits the procedural characteristics of Perlin noise to simulate realistic atmospheric fog and iteratively optimize it for effective adversarial perturbation. The overall framework consists of two major components: a procedural fog simulation module and a gradient-guided fog adversarial optimization module.

Given a clean RS image as input, the proposed FogFool method first constructs a procedural fog representation using Perlin noise. In particular, the image is partitioned into a grid, forming a grid-partitioned image, with random gradient vectors assigned to each grid vertex, yielding a gradient lattice. Based on this lattice, multiple Perlin noise maps are generated for different combinations of persistence and lacunarity, yielding six noise maps corresponding to different octaves. These multiscale noise maps are then fused using fractional Brownian motion (FBM) to produce an initial fog-intensity map, which is used to initialize the fog mask $\mathbf{C}^{(0)}$.[1] This process enables the initialization of spatially coherent, multiscale fog-like patterns that resemble real atmospheric phenomena.

In the adversarial optimization stage, the fog mask $\mathbf{C}$ is further smoothed via Gaussian filtering to obtain a smoothed fog mask, which is then blended with a predefined white color base to construct the fog layer $\mathbf{F}$ and further fused with the original RS image to generate a fog-based adversarial example $\mathbf{x}^{adv}$. The adversarial image is then fed into the target classification network, and the loss gradient with respect to the fog mask is computed. Guided by the model gradients, the fog mask is iteratively refined to maximize attack effectiveness while maintaining visual plausibility. After multiple iterations, the optimized fog mask yields adversarial examples that can successfully mislead the classification model.

To avoid ambiguity, we consistently use "fog mask" to refer to the optimizable atmospheric mask $\mathbf{C}$, "initial fog-intensity map" to refer to the FBM-based initialization $\mathbf{C}^{(0)}$, and "fog

[1]The fusion of multioctave Perlin noise via FBM is designed to mimic the multiscale fractal characteristics of real atmospheric fog, which is more physically plausible than single-scale noise-based fog simulation and avoids artificial grid artifacts common in conventional perturbation methods.

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layer" to refer to $\mathbf{F}$ obtained by blending the fog mask with the white color base.

B. Problem Description

Let $f(\cdot)$ denote a trained image classification model that maps an input RS image $\mathbf{x} \in \mathbb{R}^{H\times W\times C}$ to a probability distribution over K classes, i.e., $f(\mathbf{x}) = \mathbf{p} \in [0,1]^K$, where $\sum_{k=1}^{K} p_k = 1$. The predicted label is given by $\hat{y} = \arg\max_k f_k(\mathbf{x})$.

Given a clean image $\mathbf{x}$ with the ground-truth label y, the objective of the adversarial attack is to generate an adversarial example $\mathbf{x}^{adv}$ such that the model prediction is altered, while the applied perturbation remains visually imperceptible or physically plausible.

1) Untargeted Adversarial Attack: In an untargeted attack, the goal is to cause misclassification without specifying a target class, which can be formulated as follows:

$$\hat{y}^{adv} = \arg\max_k f_k\left(\mathbf{x}^{adv}\right), \quad \text{s.t. } \hat{y}^{adv} \neq y. \tag{1}$$

This objective can be equivalently expressed by maximizing the classification loss

$$\mathbf{x}^{adv} = \arg\max_{\mathbf{x}'\in\mathcal{S}} \mathcal{L}\left(f\left(\mathbf{x}'\right), y\right) \tag{2}$$

where $\mathcal{L}(\cdot)$ denotes the loss function (e.g., cross entropy) and $\mathcal{S}$ represents the feasible perturbation space.

2) Targeted Adversarial Attack: In a targeted attack, the adversary aims to force the model to predict a specific target label $y_t \neq y$, i.e., $\hat{y}^{adv} = y_t$. Accordingly, the adversarial example is obtained by minimizing the loss with respect to the target class

$$\mathbf{x}^{adv} = \arg\min_{\mathbf{x}'\in\mathcal{S}} \mathcal{L}\left(f\left(\mathbf{x}'\right), y_t\right). \tag{3}$$

In conventional adversarial attacks, the feasible perturbation space $\mathcal{S}$ is usually defined by an explicit ℓ_p-norm constraint, such as $\mathcal{S}_p = \{\mathbf{x}' \mid \|\mathbf{x}' - \mathbf{x}\|_p \leq \epsilon, \mathbf{x}' \in [0,1]^{H\times W\times C}\}$, where ϵ controls the maximum pixelwise perturbation magnitude. Although such a constraint limits the numerical difference between $\mathbf{x}$ and $\mathbf{x}'$, it does not explicitly encode the spatial structure or physical semantics of the perturbation.

In this work, we instead define the feasible space as a fog-constrained atmospheric transformation space. In particular, a feasible adversarial example should be generated by blending the clean image with a structured fog layer

$$\mathcal{S}_{\text{fog}} = \Big\{\mathbf{x}' \,\Big|\, \mathbf{x}' = \lambda_b\mathbf{F} + (1-\lambda_b)\,\mathbf{x}, \mathbf{F} = (1-\lambda_w)\,\mathbf{C} + \lambda_w\mathbf{1} \quad \mathbf{C} \in [0,1]^{H\times W\times C}\Big\} \tag{4}$$

where $\mathbf{C}$ is the optimizable fog mask, $\mathbf{F}$ is the fog layer, λ_w controls the fog whiteness intensity, and λ_b controls the fog blending strength. Therefore, unlike conventional ℓ_p-bounded attacks that directly perturb individual pixels, FogFool constrains adversarial examples to be generated from spatially coherent fog-like patterns. This formulation makes the perturbation space more structured and atmospherically meaningful, while the detailed construction and optimization of $\mathbf{C}$ are introduced in Sections III-C and III-D.

C. Perlin Noise

To simulate realistic atmospheric fog patterns, we adopt Perlin noise as the fundamental procedural modeling technique. Originally introduced by Perlin [38] and later refined in [39], Perlin noise is a gradient-based noise function that generates spatially coherent and continuous random fields. Due to its ability to produce natural-looking textures with controllable smoothness and scale, Perlin noise has been widely used in computer graphics to model natural phenomena, such as clouds, smoke, and fog.

In this work, Perlin noise is employed to construct a physically plausible fog-intensity distribution for remote-sensing imagery. The noise generation process consists of three main steps: grid partition and gradient lattice construction, gradient influence computation at arbitrary query points, and smooth interpolation for noise evaluation. To capture the inherent multiscale characteristics of atmospheric fog, multiple Perlin noise maps are aggregated using FBM. The detailed formulation of each step is described as follows.

Step 1: The process begins with the original RS imagery, as displayed in Fig. 3(a). The image domain is partitioned into a square lattice with dimensions $n \times n$, overlaying the grid structure shown in Fig. 3(b). Subsequently, a random 2-D gradient vector is allocated to each node (intersection) of the grid, establishing the noise basis visualized in the schematic grid in Fig. 3(c).

Step 2: Let $P = \{p_{x,y} | 0 \leq x, y < n\}$ be the set of query points. For an arbitrary point $p_{x,y}$ [located within the grid in Fig. 3(c)], the surrounding unit square is delimited by four vertices $V = \{v_{i,j} | i = l, r; j = d, u\}$. The coordinate bounds are calculated as $l = \lfloor x \rfloor$ and $d = \lfloor y \rfloor$, with $r = l + 1$ and $u = d + 1$, where $\lfloor\cdot\rfloor$ denotes the floor operation. As detailed in the magnified view in Fig. 3(d), let $\vec{g}_{i,j}$ denote the random gradient vectors at the vertices. The displacement vector, representing the offset from a grid corner $v_{i,j}$ to the target point $p_{x,y}$, is expressed as follows:

$$\vec{d}_{i,j} = \left[x - i, y - j\right]. \tag{5}$$

Finally, the scalar influence $s_{i,j}$ (gradient ramp value) is derived for each vertex via the dot product of the displacement and gradient vectors

$$s_{i,j} = \vec{d}_{i,j} \cdot \vec{g}_{i,j}. \tag{6}$$

Step 3: After obtaining the gradient ramp values $s_{i,j}$ at the four surrounding vertices, the Perlin noise value at an arbitrary query point $p_{x,y}$ is computed through a smooth interpolation process. To ensure continuity and suppress grid artifacts, a quintic smoothing function, commonly referred to as the fade function, was proposed in [39] and is applied to the normalized local coordinates

$$\text{fade}\,(t) = 6t^5 - 15t^4 + 10t^3. \tag{7}$$

Let $\Delta x = x - l$ and $\Delta y = y - d$ denote the relative offsets of the query point within the unit grid cell. The fade function is then evaluated at Δx and Δy to produce smooth interpolation weights. Subsequently, linear interpolation is utilized to blend the gradient values smoothly, where

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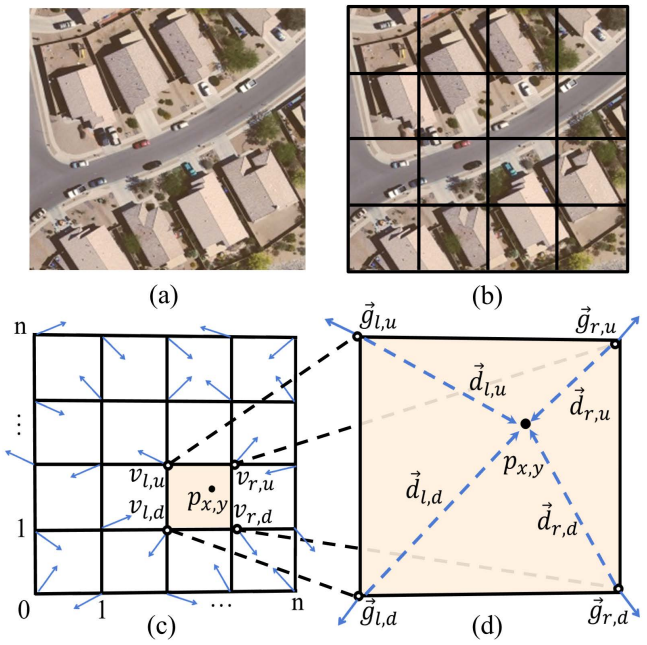


Fig. 3. Process of 2-D Perlin noise generation for fog-mask initialization. (a) Original RS image. (b) Image is partitioned into an $n \times n$ grid. (c) Random gradient vectors are assigned to grid vertices. (d) Noise value at a query point is computed from the dot products between the gradient vectors $\vec{g}$ and displacement vectors $\vec{d}$, followed by smooth interpolation.

the function $\text{lerp}(a, b, t)$ serves as a fundamental linear interpolation operator. Formally, it is defined as follows:

$$\text{lerp}\,(a, b, t) = a + t\,(b - a)\,. \tag{8}$$

First, the gradient ramp values along the horizontal direction are interpolated at the lower and upper edges of the grid cell

$$\begin{aligned} l_d &= \text{lerp}\left(s_{l,d}, s_{r,d}, \text{fade}\,(\Delta x)\right) \\ l_u &= \text{lerp}\left(s_{l,u}, s_{r,u}, \text{fade}\,(\Delta x)\right). \end{aligned} \tag{9}$$

Subsequently, a second interpolation is carried out along the vertical direction to obtain the final Perlin noise value at $p_{x,y}$

$$f\,(x, y) = \text{lerp}\,(l_d, l_u, \text{fade}\,(\Delta y))\,. \tag{10}$$

Through this two-stage interpolation process, the resulting noise field is continuous and smooth across grid boundaries, forming the basic Perlin noise map used in the subsequent multioctave fog modeling.

While a single-layer Perlin noise produces spatially smooth but relatively simple patterns, realistic atmospheric fog exhibits multiscale structures with both coarse and fine variations. To capture such characteristics, multiple Perlin noise maps at different spatial frequencies are combined using FBM.

Formally, the FBM noise field is defined as a weighted summation of Perlin noise functions across multiple octaves

$$\text{FBM}\,(x, y) = \sum_{k=0}^{K-1} \alpha_k \cdot f\,(\beta_k x, \beta_k y) \tag{11}$$

where $f(\cdot)$ denotes the base Perlin noise function defined in (10), K represents the total number of octaves, α_k is the amplitude scaling factor (persistence), and β_k is the frequency scaling factor (lacunarity) at the kth octave.

In this work, we adopt a commonly used exponential scaling strategy for FBM to balance visual realism and computational efficiency, following the standard Perlin noise configuration [38]. The number of octaves is set to $K = 6$ to sufficiently model the multiscale structure of natural fog without incurring excessive computational cost. The amplitude factor $\alpha_k = 2^{-k}$ progressively weakens high-frequency fine details, consistent with the natural attenuation of atmospheric features, while the frequency factor $\beta_k = 2^k$ doubles the spatial frequency per octave to generate spatially coherent, realistic fog patterns.

By aggregating Perlin noise maps at increasing frequencies and decreasing amplitudes, the resulting FBM noise field exhibits scale-invariant, spatially coherent patterns that closely resemble real atmospheric fog. The generated FBM map is used as the initial fog-intensity map $\mathbf{C}^{(0)}$ for subsequent fog-mask optimization.

D. Fog Adversarial Example Optimization

After initializing the fog mask with the FBM-based fog-intensity map, we further optimize the fog mask to generate effective fog-based adversarial examples. Unlike conventional pixelwise perturbations (e.g., FGSM [17], BIM [18], PGD [19], and MI-FGSM [20]), FogFool restricts the perturbation space to a structured fog mask, thereby preserving physical plausibility while enabling adversarial manipulation.

1) Fog-Based Image Formation: Let $\mathbf{x} \in [0, 1]^{H \times W \times C}$ denote the clean RS image, and let $\mathbf{C} \in [0, 1]^{H \times W \times C}$ denote the optimizable fog mask initialized by the FBM-based fog-intensity map defined in (11).

To simulate atmospheric scattering, we construct a fog layer by blending the fog mask with a white color base

$$\mathbf{F} = (1 - \lambda_w)\,\mathbf{C} + \lambda_w \mathbf{1} \tag{12}$$

where $\lambda_w \in [0, 1]$ controls the fog whiteness intensity and $\mathbf{1}$ denotes an all-one tensor.

The adversarial image is then obtained via linear blending

$$\mathbf{x}^{adv} = \lambda_b \mathbf{F} + (1 - \lambda_b)\,\mathbf{x} \tag{13}$$

where $\lambda_b \in [0, 1]$ controls the fog blending strength. The resulting image is clipped to ensure that pixel values fall within valid ranges.

2) Gradient-Guided Fog Optimization: The fog mask $\mathbf{C}$ is treated as the optimization variable. The objective is defined according to the adversarial attack setting.

For untargeted attacks, we maximize the classification loss

$$\max_{\mathbf{C} \in [0,1]} \mathcal{L}\left(f\left(\mathbf{x}^{adv}\right), y\right) \tag{14}$$

while for targeted attacks, we minimize

$$\min_{\mathbf{C} \in [0,1]} \mathcal{L}\left(f\left(\mathbf{x}^{adv}\right), y_t\right). \tag{15}$$

To solve this optimization problem, we adopt an iterative gradient-sign strategy inspired by the momentum-based iterative FGSM. At iteration t, the gradient with respect to the fog mask is computed as follows:

$$\mathbf{g}^{(t)} = \nabla_{\mathbf{C}^{(t)}} \mathcal{L}\left(f\left(\mathbf{x}^{adv}\right), y\right). \tag{16}$$

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Algorithm 1 Fog Adversarial Example Optimization

Input: Clean image $\mathbf{x}$, label y, classifier $f(\cdot)$
Output: Adversarial example $\mathbf{x}^{adv}$
Initialize fog mask $\mathbf{C}^{(0)}$ using the FBM-based fog-intensity map;
Initialize momentum $\mathbf{m}^{(0)} = 0$;
for $t = 0$ *to* $T - 1$ **do**
  Construct fog layer $\mathbf{F}$ using Eq. (12);
  Generate adversarial image $\mathbf{x}^{adv}$ using Eq. (13);
  Compute loss $\mathcal{L}$ according to attack setting;
  Compute gradient $\mathbf{g}^{(t)} = \nabla_{\mathbf{C}^{(t)}}\mathcal{L}$;
  Normalize gradient;
  Update momentum using Eq. (18);
  Update fog mask using Eq. (19);
  Apply Gaussian smoothing using Eq. (20);
Construct final adversarial image using optimized fog mask;
return $\mathbf{x}^{adv}$;

The gradient is first normalized by its mean absolute value

$$\tilde{\mathbf{g}}^{(t)} = \frac{\mathbf{g}^{(t)}}{\text{mean}\left(|\mathbf{g}^{(t)}|\right)}. \tag{17}$$

To stabilize optimization and improve transferability, a momentum term is incorporated

$$\mathbf{m}^{(t)} = \mu\mathbf{m}^{(t-1)} + \tilde{\mathbf{g}}^{(t)} \tag{18}$$

where μ denotes the decay factor.

The fog mask is updated using the sign of the accumulated gradient for untargeted attacks (the sign is reversed for targeted attacks)

$$\mathbf{C}^{(t+1)} = \Pi_{[0,1]}\left(\mathbf{C}^{(t)} + \alpha \cdot \text{sign}\left(\mathbf{m}^{(t)}\right)\right) \tag{19}$$

where α is the step size and $\Pi_{[0,1]}$ denotes projection onto the valid intensity range.

3) Fog Naturalness Regularization: To preserve the spatial smoothness and natural appearance of fog, Gaussian filtering is applied to the updated fog mask after each iteration

$$\mathbf{C}^{(t+1)} \leftarrow \mathcal{G}_\sigma\left(\mathbf{C}^{(t+1)}\right) \tag{20}$$

where $\mathcal{G}_\sigma(\cdot)$ denotes Gaussian smoothing with standard deviation σ. After T iterations, the final adversarial example is obtained using (12) and (13) with the optimized fog mask.

4) Algorithm Summary: The overall fog adversarial optimization procedure is summarized in Algorithm 1, which mainly consists of three core steps: constructing an initial fog intensity map via multioctave Perlin noise, performing gradient-guided optimization to adjust the fog map for effective adversarial attacks while retaining natural fog characteristics, and iteratively updating and verifying until the termination condition is satisfied. The algorithm has a linear time complexity $O(H \times W \times C)$ (adaptive to input image size) due to fixed parameters K and T, and it is highly practical with no complex preprocessing requirements, controllable parameters, and efficient generation of adversarial examples for evaluating RS classifiers in digital processing pipelines.

IV. Experiments

To comprehensively evaluate the effectiveness of the proposed method, we conduct both untargeted and targeted adversarial attack experiments on two widely used RS scene classification datasets: UCM [40] and NWPU-RESISC45 (NWPU) [41].

A. Datasets and Comparison Methods

1) Datasets:

UCM: The UCM dataset consists of 21 land-use scene categories, with 100 images per class. Each image has a spatial resolution of 256×256 pixels and a ground sampling distance of approximately 1 ft. The dataset was collected from the United States Geological Survey (USGS) National Map Urban Area Imagery Collection, covering diverse urban regions.

NWPU-RESISC45 (NWPU): The NWPU dataset is a large-scale benchmark for RS scene classification, comprising 31 500 images across 45 scene categories, with 700 images per category. To ensure a fair evaluation of the proposed FogFool, the "cloud" category is excluded to avoid semantic overlap with the simulated fog perturbations. Consequently, 44 categories comprising 30 800 images are used in our experiments.

2) Target Models: For both datasets, we adopt a stratified split, allocating 80% of the samples from each class to training and the remaining 20% to testing. To comprehensively evaluate the effectiveness of FogFool across different architectures, we select eight widely adopted CNN models in RS image classification, including AlexNet [42], VGG16 [43], ResNet50 and ResNet101 [44], DenseNet121 and DenseNet201 [45], MobileNetV2 [46], and EfficientNet-B0 [47]. In addition to CNN-based models, we further include two representative Transformer-based image classification models, namely ViT-B/16 [48] and Swin Transformer-Tiny (Swin-T) [49], to evaluate the effectiveness of FogFool on nonconvolutional architectures. The experimental platform is based on Ubuntu 22.04.5 LTS and the PyTorch framework [50], with an Intel Xeon Gold 6342 24-Core processor CPU and an NVIDIA RTX A40 GPU. Each model is trained separately on the UCM and NWPU datasets based on pretrained weights. The batch size is 32, and the learning rate is 10^{-4}. The AdamW optimizer is employed with a weight decay of 10^{-4} for 50 training epochs. The input size of the images is 224×224, and data augmentation operations, e.g., random vertical or horizontal flips, are used during training. The test accuracy of each model on the two datasets is shown in Table I.

3) Comparison Methods: To comprehensively evaluate the effectiveness of FogFool, we compare it with seven representative white-box attack methods, including FGSM [17], BIM [18], C&W [16], PGD [19], MI-FGSM [20], Jitter [51], and AutoAttack [52]. To further assess transferability in the black-box setting, we adopt seven widely used transfer-based attacks, including MI-FGSM [20], DI-FGSM [53], TI-FGSM [54], NI-FGSM [55], SI-NI-FGSM [55], VMI-FGSM [56], and VNI-FGSM [56].

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TABLE I
TEST ACCURACY (%) OF DIFFERENT MODELS ON UCM AND NWPU DATASETS

Dataset	AlexNet	VGG16	ResNet50	ResNet101	DenseNet121	DenseNet201	MobileNetV2	EfficientNet-B0	ViT-B/16	Swin-T
UCM	88.33	93.33	91.67	95.24	94.05	93.10	90.71	92.62	95.23	96.14
NWPU	90.89	93.10	95.52	95.99	95.73	96.17	95.03	95.99	96.19	96.72

4) Metrics: To quantitatively evaluate the attack performance, we adopt the attack success rate (ASR) as the primary evaluation metric, defined as follows:

$$\mathrm{ASR} = \frac{n_{\mathrm{adv}}}{n_{\mathrm{total}} - n_{\mathrm{mis}}} \tag{21}$$

where n_{total} denotes the total number of test samples and n_{mis} represents the number of samples that are already misclassified by the model. These misclassified samples are excluded from evaluation because adversarial perturbations are unnecessary for them. n_{adv} denotes the number of successfully generated adversarial examples that cause the target model to change its prediction. For targeted attacks, ASR measures the proportion of adversarial examples that are successfully classified into the specified target class. A higher ASR indicates stronger attack capability under the same perturbation constraint.

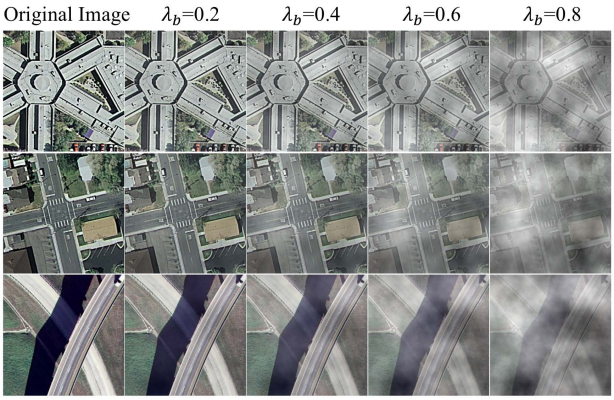


Fig. 4. Illustration of fog-based adversarial examples under different fog blending coefficients λ_b. From (left to right) clean image, $\lambda_b = 0.2$, $\lambda_b = 0.4$, $\lambda_b = 0.6$, and $\lambda_b = 0.8$. Increasing λ_b intensifies the fog effect and increases the magnitude of the structured perturbation.

B. Experimental Settings of the Proposed Method

In this section, we present the implementation details and parameter configurations of FogFool. Unlike conventional adversarial attacks that constrain perturbations under a predefined ℓ_p-norm bound, FogFool restricts perturbations to an optimizable fog mask initialized by an FBM-based fog-intensity map. No explicit perturbation budget (e.g., ϵ under ℓ_∞-norm) is imposed. Instead, the perturbation strength is implicitly controlled via fog-blending parameters and structured-mask optimization.

1) Fog Initialization: The fog mask $\mathbf{C}^{(0)}$ is initialized by the FBM-based fog-intensity map constructed from multioctave Perlin noise. The number of octaves is set to $K = 6$. The amplitude and frequency scaling factors follow exponential schedules defined as $\alpha_k = 2^{-k}$ and $\beta_k = 2^k$, respectively.

2) Fog Formation Parameters: The fog whiteness coefficient in (12) is set to $\lambda_w = 0.2$, which controls the intensity of atmospheric scattering. The fog blending coefficient in (13) is set to $\lambda_b = 0.6$, balancing visibility preservation and attack strength. As demonstrated in Fig. 4, varying λ_b produces progressively denser fog effects, demonstrating its direct influence on the perceptual strength of the generated adversarial examples.

3) Optimization Settings: The fog mask is optimized for $T = 20$ iterations using a gradient-sign-update strategy with momentum. The step size is set to $\alpha = 1/255$, and the momentum decay factor is $\mu = 1.0$. To preserve spatial smoothness and enhance perceptual realism, Gaussian smoothing with standard deviation $\sigma = 0.7$ is applied after each update. All parameter settings of the proposed method are summarized in Table II.

TABLE II
PARAMETER SETTINGS OF THE PROPOSED FOG-BASED ATTACK

Parameter	K	α_k	β_k	λ_w	λ_b	T	α	μ	σ
Value	6	2^{-k}	2^k	0.2	0.6	20	1/255	1.0	0.7

C. Experimental Results and Analysis

1) Parameter Analysis: To achieve a proper balance between attack effectiveness and visual plausibility, we conduct a systematic analysis of two key hyperparameters in the FogFool: the fog blending coefficient λ_b and the number of optimization iterations T.

Fig. 5 presents the classification accuracy of all evaluated models on the UCM and NWPU datasets as λ_b increases from 0 to 0.9, without applying adversarial optimization. When $\lambda_b = 0$, no fog blending is introduced, and the models maintain their baseline accuracy, consistent with the results reported in Table I. As λ_b increases, the fog intensity becomes progressively stronger, resulting in a gradual decline in classification accuracy. When $\lambda_b \leq 0.6$, the accuracy degradation remains limited, with most models preserving over 90% of their clean performance. Within this range, the fog perturbation remains visually plausible and does not substantially distort the semantic content of the RS scenes. On the contrary, when $\lambda_b > 0.6$, the accuracy drops significantly across models, indicating that excessive fog blending severely obscures discriminative features and compromises scene recognizability. Based on these observations, we adopt $\lambda_b = 0.6$ as the default setting. This choice provides a favorable tradeoff, maintaining perceptual realism while preserving sufficient perturbation capacity for subsequent adversarial optimization.

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TABLE III
ASRs (%) OF DIFFERENT METHODS ON THE UCM DATASET (UNTARGETED ATTACK)

Attack method	AlexNet	VGG16	ResNet50	ResNet101	DenseNet121	DenseNet201	MobileNetV2	EfficientNet-B0	Average
FGSM	90.84	74.49	43.64	56.00	57.72	60.61	57.48	82.26	65.38
BIM	100.00	98.98	100.00	100.00	99.75	100.00	99.21	100.00	99.74
CW	79.78	90.05	95.32	97.00	95.95	96.93	94.49	96.92	93.31
PGD	100.00	98.72	100.00	100.00	100.00	100.00	99.48	100.00	99.77
MIFGSM	100.00	98.47	100.00	99.50	99.49	100.00	99.74	100.00	99.65
Jitter	90.84	82.65	95.32	94.75	95.95	94.88	97.90	99.49	93.97
AutoAttack	100.00	100.00	100.00	100.00	100.00	100.00	99.74	98.71	**99.81**
FogFool	98.11	93.62	96.62	96.25	96.46	98.98	97.90	96.40	96.79

TABLE IV
ASRs (%) OF DIFFERENT METHODS ON THE UCM DATASET (TARGETED ATTACK)

Attack method	AlexNet	VGG16	ResNet50	ResNet101	DenseNet121	DenseNet201	MobileNetV2	EfficientNet-B0	Average
BIM	97.57	95.66	98.70	99.00	99.49	99.49	98.43	99.23	98.45
PGD	96.23	94.64	98.18	98.25	99.49	99.74	98.69	98.71	97.99
MIFGSM	97.84	97.45	100.00	99.50	99.49	99.74	98.43	100.00	**99.06**
FogFool	96.23	92.35	98.18	99.25	99.49	100.00	97.38	98.46	97.67

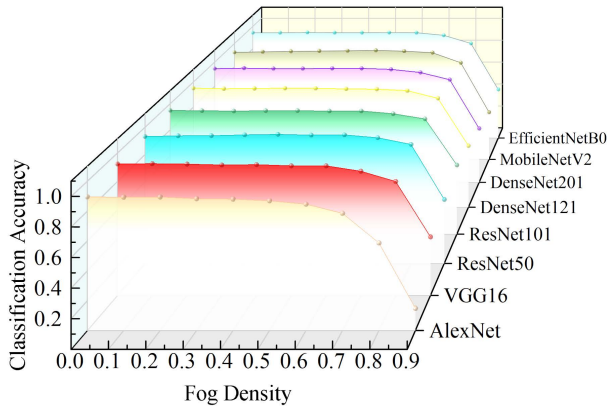


Fig. 5. Classification accuracy of all evaluated models as a function of fog density λ_b, averaged across the UCM and NWPU datasets. Fog is applied without adversarial optimization to simulate natural atmospheric interference.

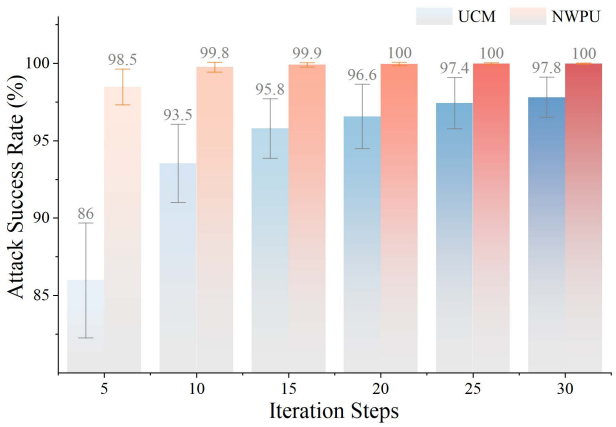


Fig. 6. ASR versus the number of optimization iterations T with λ_b fixed at 0.6. Results are averaged across all evaluated models, with blue bars representing the UCM dataset and red bars representing the NWPU dataset. Error bars indicate the standard deviation of ASRs across different models.

Fig. 6 illustrates the ASR of FogFool as a function of the number of optimization iterations T, with λ_b fixed at 0.6. On both datasets, the ASR increases rapidly during the initial iterations, demonstrating that the gradient-guided optimization effectively refines the structured fog mask to induce misclassification. As T increases, the ASR gradually saturates, and the performance gain becomes marginal beyond approximately 20 iterations. This convergence pattern is consistently observed across different model architectures, indicating that the optimization process stabilizes within a relatively small number of update steps. Considering both attack performance and computational efficiency, we set the number of iterations to $T = 20$ in subsequent experiments. This configuration achieves near-saturated ASR while keeping the computational overhead moderate.

2) Quantitative Analysis: We proceed to comprehensively evaluate the attack performance of the proposed FogFool method in both untargeted and targeted settings. The quantitative results are summarized in Tables III–VI, and representative adversarial examples are illustrated in Figs. 7 and 8. As shown in Table III, FogFool achieves an average ASR of 96.79% on the UCM dataset, maintaining consistent performance across all eight models (93.62%–98.98%). On the NWPU dataset (Table V), FogFool reaches an average ASR of 99.96%, comparable to the state-of-the-art methods (PGD: 99.97%, AutoAttack: 100.00%), with 100% ASR on six models, demonstrating effective untargeted attack capability.

Tables IV and VI present targeted attack results. On the UCM dataset, FogFool achieves an average ASR of 97.67%, with a maximum of 100.00% on DenseNet201. On the NWPU dataset, it reaches 99.93% average ASR, only 0.04% lower than the top-performing MI-FGSM, confirming qualified targeted attack performance.

Figs. 7 and 8 show that FogFool generates visually natural adversarial examples. The Perlin noise-based fog masks are spatially coherent and realistic, and the adversarial samples

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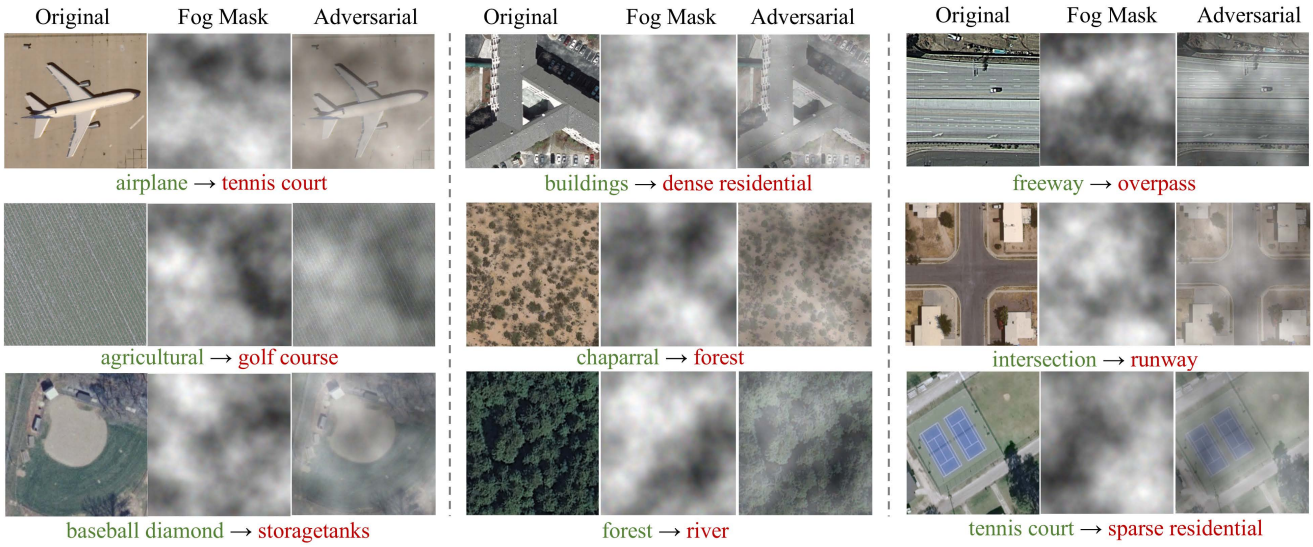


Fig. 7. Adversarial examples generated by the proposed method on the UCM dataset. Each group of images shows, from (left to right), the original image, the optimized fog mask, and the corresponding fog-based adversarial example. Green labels indicate the correct categories, and red labels indicate the labels predicted by the target models.

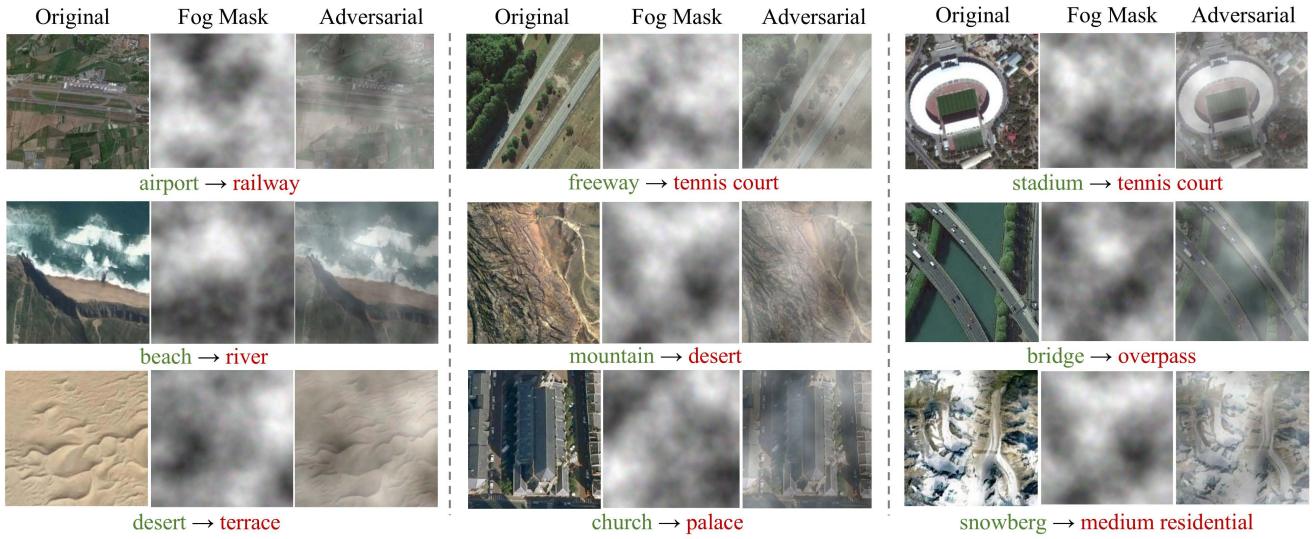


Fig. 8. Adversarial examples generated by the proposed method on the NWPU dataset. Each group of images shows, from (left to right), the original image, the optimized fog mask, and the corresponding fog-based adversarial example. Green labels indicate the correct categories, and red labels indicate the labels predicted by the target models.

TABLE V

ASRs (%) OF DIFFERENT METHODS ON THE NWPU DATASET (UNTARGETED ATTACK)

Attack method	AlexNet	VGG16	ResNet50	ResNet101	DenseNet121	DenseNet201	MobileNetV2	EfficientNet-B0	Average
FGSM	91.12	87.20	73.28	80.33	84.25	83.79	82.30	82.21	83.06
BIM	99.95	99.69	100.00	100.00	100.00	100.00	100.00	100.00	99.95
CW	84.10	88.88	86.15	83.58	93.35	90.61	96.24	96.36	89.91
PGD	99.96	99.81	100.00	100.00	100.00	100.00	100.00	100.00	99.97
MIFGSM	99.95	99.69	100.00	100.00	100.00	100.00	100.00	100.00	99.95
Jitter	99.96	99.84	98.88	99.04	98.34	97.79	99.71	99.90	99.18
AutoAttack	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	**100.00**
FogFool	99.96	99.72	100.00	100.00	100.00	100.00	100.00	100.00	99.96

maintain physical plausibility without obvious artifacts, while effectively misleading target models.

Overall, the experimental results demonstrate that FogFool achieves competitive ASR under both untargeted and targeted settings across two benchmark RS datasets. Importantly, these results are obtained without relying on explicit ℓ_p-norm perturbation constraints. Instead, the attack operates within a physically interpretable fog generation framework, offering a

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TABLE VI
ASRs (%) OF DIFFERENT METHODS ON THE NWPU DATASET (TARGETED ATTACK)

Attack method	AlexNet	VGG16	ResNet50	ResNet101	DenseNet121	DenseNet201	MobileNetV2	EfficientNet-B0	Average
BIM	99.48	99.18	99.88	99.81	100.00	100.00	100.00	99.97	99.79
PGD	99.46	99.13	99.90	99.85	100.00	100.00	100.00	99.98	99.79
MIFGSM	99.93	99.84	100.00	100.00	100.00	100.00	100.00	100.00	**99.97**
FogFool	99.96	99.51	100.00	100.00	100.00	100.00	100.00	100.00	99.93

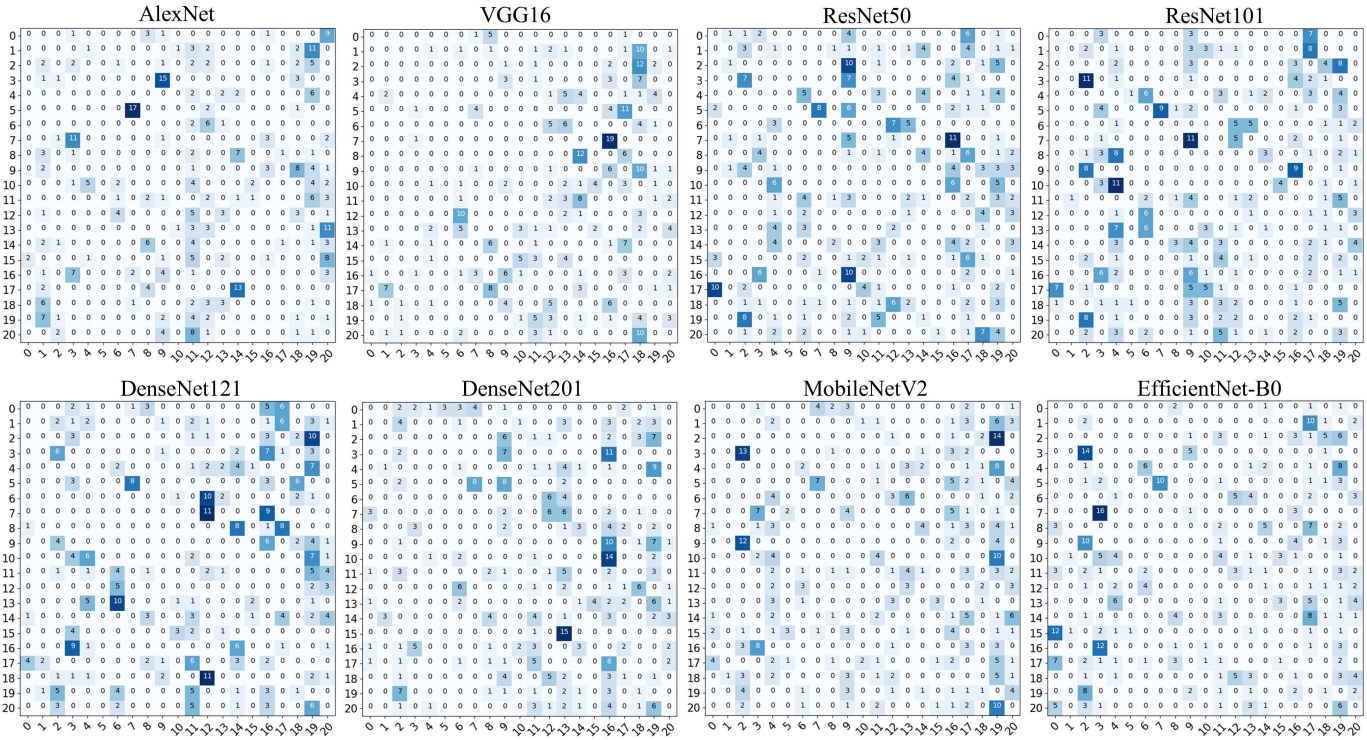


Fig. 9. Confusion matrices of fog adversarial examples on the UCM dataset for different target models. Rows represent ground-truth labels, and columns represent predicted labels. The label numbers from 0 to 20 represent the categories as follows: agricultural, airplane, baseball diamond, beach, buildings, chaparral, dense residential, forest, freeway, golf course, harbor, intersection, medium residential, mobile home park, overpass, parking lot, river, runway, sparse residential, storage tanks, and tennis court.

favorable balance between attack strength and visual realism. This highlights the potential vulnerability of RS models to semantically meaningful and physically plausible adversarial manipulations.

3) Attack Selectivity: Chen et al. [57] introduced the concept of *attack selectivity* in RS image classification, showing that adversarial misclassifications are not uniformly distributed across categories but instead concentrate on specific target classes. To further investigate this phenomenon, Fig. 9 presents the confusion matrices of fog adversarial examples on the UCM dataset.

Notably, the selective misclassification behavior varies across model architectures. For instance, ResNet50 exhibits a strong tendency to misclassify adversarial examples into category 9 (golf course), whereas DenseNet201 shows a clear bias toward category 16 (river). This suggests that attack selectivity is closely related to the geometric properties of the decision boundaries learned by different neural network architectures. Under fog-induced perturbations, feature representations may be systematically shifted toward particular regions of the classification space.

Furthermore, intrinsic semantic similarity between categories also plays a crucial role. Regardless of the target model, samples from category 5 (chaparral) are predominantly misclassified as category 7 (forest). Given the high visual and semantic similarity between these land-cover types, their corresponding feature embeddings are likely to be closely distributed in high-dimensional representation space. Consequently, small structured perturbations can more easily push such samples across decision boundaries into semantically adjacent categories.

4) Transfer Attack: To evaluate the cross-model transferability of the FogFool, we conduct ensemble-based transfer attack experiments. In particular, three models (i.e., ResNet50, DenseNet121, and MobileNet_V2) are selected as surrogate models to generate adversarial examples in a white-box setting. The generated samples are then directly applied to attack five unseen target models in the black-box setting. The transfer ASRs (TASRs) on the UCM and NWPU datasets are reported in Tables VII and VIII, respectively.

As shown in Table VII, FogFool achieves an average TASR of 51.58% on the UCM dataset, outperforming all compared

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TABLE VII
ENSEMBLE-BASED TASRS (%) ON THE UCM DATASET

Attack method	ResNet50	DenseNet121	MobileNetV2	AlexNet	VGG16	ResNet101	DenseNet201	EfficientNet-B0	Average
MI-FGSM	100.00	95.70	98.16	12.40	24.49	39.50	36.06	62.98	35.08
DI-FGSM	96.88	94.18	96.85	15.09	35.20	60.75	54.99	67.87	46.78
TI-FGSM	98.18	94.68	95.54	19.41	40.56	61.75	62.40	68.89	50.60
NI-FGSM	98.44	94.94	98.16	11.86	23.21	38.00	34.27	64.78	34.42
SI-NI-FGSM	97.14	95.70	97.38	13.21	28.32	47.25	39.64	64.78	38.64
VMI-FGSM	100.00	95.44	98.69	15.90	34.95	59.75	56.01	69.67	47.25
VNI-FGSM	100.00	96.20	98.16	15.63	35.97	62.75	57.29	70.95	48.51
FogFool	92.99	91.65	91.60	44.20	41.84	56.75	56.52	58.61	**51.58**

TABLE VIII
ENSEMBLE-BASED TASRS (%) ON THE NWPU DATASET

Attack method	ResNet50	DenseNet121	MobileNetV2	AlexNet	VGG16	ResNet101	DenseNet201	EfficientNet-B0	Average
MI-FGSM	99.97	99.86	100.00	26.90	49.78	72.87	73.21	62.74	57.10
DI-FGSM	99.93	99.61	100.00	37.61	69.08	86.34	85.16	81.04	71.84
TI-FGSM	99.86	99.61	100.00	48.44	73.67	86.96	87.59	84.63	76.25
NI-FGSM	99.93	99.83	100.00	25.63	48.20	69.64	67.59	59.83	54.17
SI-NI-FGSM	98.88	99.19	100.00	29.34	52.17	70.89	67.00	64.06	56.69
VMI-FGSM	99.97	99.66	100.00	37.19	66.70	84.56	83.19	79.00	70.12
VNI-FGSM	99.98	99.76	100.00	38.13	70.78	87.13	85.25	80.65	72.38
FogFool	99.63	99.71	99.93	74.64	76.13	89.13	91.31	87.49	**83.74**

TABLE IX
AVERAGE LINEAR CKA VALUES OF FEATURE DISCREPANCY BETWEEN SURROGATE MODELS AND TARGET MODELS

Dataset	Attack Method	AlexNet	VGG16	ResNet101	DenseNet201	EfficientNet-B0	Average
UCM	MI-FGSM	0.1243	0.0298	0.3865	0.4335	0.3179	0.2584
	DI-FGSM	0.1812	0.0722	0.5886	0.5756	0.4553	0.3746
	TI-FGSM	0.1900	0.0781	0.5857	0.5727	0.4800	0.3813
	NI-FGSM	0.1372	0.0296	0.5617	0.5191	0.3244	0.3144
	SI-NI-FGSM	0.1738	0.0365	0.5831	0.6179	0.4541	0.3731
	VMI-FGSM	0.1449	0.0443	0.4682	0.5000	0.4356	0.3186
	VNI-FGSM	0.1724	0.0740	0.6229	0.5983	0.5034	0.3942
	FogFool	0.2245	0.0458	0.6070	0.5657	0.5291	**0.3944**
NWPU	MI-FGSM	0.1791	0.0831	0.4363	0.4308	0.2836	0.2826
	DI-FGSM	0.2080	0.1065	0.5721	0.5340	0.3610	0.3563
	TI-FGSM	0.2168	0.1110	0.5768	0.5882	0.4189	0.3823
	NI-FGSM	0.1951	0.1130	0.4761	0.4445	0.3043	0.3066
	SI-NI-FGSM	0.2255	0.1727	0.5509	0.4952	0.3758	0.3640
	VMI-FGSM	0.2083	0.1210	0.5420	0.5383	0.3855	0.3590
	VNI-FGSM	0.2235	0.1291	0.5634	0.5538	0.4040	0.3748
	FogFool	0.2102	0.1053	0.6527	0.6498	0.6273	**0.4490**

gradient-based ensemble methods. Although the ASRs on the three surrogate models are slightly lower than those of iterative gradient-based attacks, FogFool exhibits substantially improved transferability on the five unseen target models. In particular, the TASRs on AlexNet and VGG16 are noticeably higher than those of the other methods. This suggests that the proposed fog-based perturbations using FogFool do not overly adapt to the surrogate models' gradients. Instead, they induce more globally effective feature shifts that generalize across heterogeneous architectures, thereby enhancing cross-model transferability.

On the larger and more complex NWPU dataset (Table VIII), the superiority of the proposed FogFool becomes even more evident. FogFool achieves an average TASR of 83.74%, significantly exceeding all baseline attacks. Moreover, the TASRs on all five unseen target models consistently exceed 74%, reaching up to 91.31% on DenseNet201. Compared with conventional gradient-based ensemble attacks, which tend to rely heavily on surrogate-specific gradient alignment, FogFool generates structured, low-frequency fog perturbations that resemble realistic atmospheric effects. Such perturbations are more likely to disrupt shared and robust features learned across different architectures, leading to stronger transferability.

To further reveal the intrinsic mechanism underlying the superior transferability of FogFool, we adopt linear centered kernel alignment (CKA) to quantify the consistency of intermediate feature deviations across models. As reported in Table IX, we first extract deep backbone features of both clean images and adversarial examples from the surrogate and target networks, and compute the feature difference between adversarial and clean representations to characterize layerwise perturbation offsets. Then, linear CKA is conducted on these

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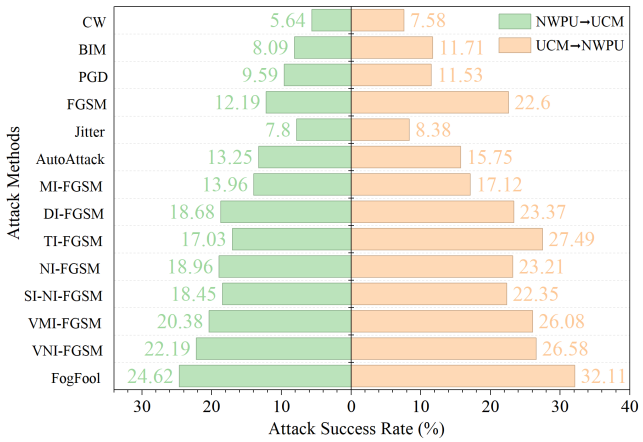


Fig. 10. Cross-dataset TASRs (%) between NWPU and UCM. For each attack method, adversarial perturbations generated on the source dataset are randomly applied to clean images from the target dataset without further optimization. Results are averaged over all ten victim models.

intermediate deviation maps between the surrogate models and black-box targets, with averaged CKA scores adopted to evaluate the similarity of cross-model feature shifting directions.

Experimental results on both UCM and NWPU datasets consistently demonstrate that FogFool achieves the highest average CKA value among all attack methods. This indicates that the fog-based adversarial examples generated by FogFool can induce much more similar intermediate feature discrepancy directions across the surrogate models and diverse target models, compared with traditional pixelwise gradient-based attacks. The structured, low-frequency, and physically plausible fog perturbations optimize the universal feature space shared by different network architectures, rather than overfitting to the specific gradient characteristics of the surrogate models. This consistent feature deviation pattern across models explains why FogFool possesses outstanding cross-model transferability in black-box RS image classification scenarios.

5) Cross-Dataset Transfer Attack: To further evaluate the transferability of adversarial perturbations beyond model architectures, we conduct cross-dataset transfer experiments between UCM and NWPU. In particular, for each attack method, adversarial examples are first generated on the source dataset using the evaluated victim models, and the corresponding adversarial perturbations are extracted. These perturbations are then randomly applied to clean images from the target dataset without further optimization. The ASRs are averaged over all ten victim models, including both CNN-based and Transformer-based architectures. This setting evaluates whether the learned perturbation patterns can generalize across datasets with different scene distributions.

As shown in Fig. 10, FogFool achieves the strongest cross-dataset transferability in both transfer directions. When transferring perturbations from NWPU to UCM, FogFool obtains an average ASR of 24.62%, outperforming the strongest baseline VNI-FGSM by 2.43 percentage points. When transferring perturbations from UCM to NWPU, FogFool achieves an average ASR of 32.11%, exceeding the best competing method TI-FGSM by 4.62 percentage points. Compared with conventional gradient-based perturbations, the fog-based perturbations generated by FogFool exhibit stronger dataset-level generalization.

This advantage can be attributed to the structured and scene-independent nature of the proposed fog-based perturbations. Instead of relying solely on dataset-specific pixel-level patterns, FogFool optimizes spatially coherent atmospheric perturbations with mid-to-low-frequency characteristics. Such perturbations are more likely to affect common visual cues shared by different RS datasets, thereby improving transferability across different data distributions.

6) Evaluation on Transformer-Based Models: To verify that FogFool is not limited to CNN-based architectures, we further evaluate it on two representative Transformer-based image classification models, including vision Transformer (ViT-B/16) [48] and Swin-T [49]. ViT-B/16 represents a standard pure Transformer architecture, while Swin-T introduces a hierarchical shifted-window attention mechanism. All Transformer-based models are trained and evaluated on the same UCM and NWPU datasets using the same data splits and attack settings as the CNN-based experiments.

As shown in Table X, FogFool remains highly effective on Transformer-based victim models. It achieves ASRs of 98.27% and 99.13% on ViT-B/16 and Swin-T for UCM, and 100.00% and 99.88% for NWPU, respectively. The average ASR reaches 99.32%, outperforming all compared methods. These results indicate that the proposed fog-based perturbations are effective not only for CNNs but also for Transformer-based classifiers.

To further examine cross-architecture transferability, we evaluate CNN-to-Transformer black-box transfer attacks. Following the previous ensemble-based transfer setting, adversarial examples are generated using ResNet50, DenseNet121, and MobileNetV2 as CNN surrogate models, and are then directly tested on unseen Transformer-based target models.

As reported in Table XI, FogFool achieves the best CNN-to-Transformer transferability, with an average TASR of 54.43%, exceeding the strongest baseline TI-FGSM by 16.47 percentage points. The improvement is particularly evident on NWPU, where FogFool obtains 69.98% and 78.72% TASR on ViT-B/16 and Swin-T, respectively. This suggests that the structured mid-to-low-frequency fog perturbations generated by FogFool are less dependent on surrogate-specific gradients and can better generalize across substantially different architectures. Overall, these additional experiments verify the effectiveness and transferability of FogFool on Transformer-based models.

7) Image Quality Assessment: To quantitatively evaluate the visual quality of adversarial examples, we further adopt peak signal-to-noise ratio (PSNR) and structural similarity index measure (SSIM) as complementary image quality metrics. The compared baselines are representative transfer-based attack methods. PSNR measures the pixel-level distortion between clean and adversarial images, while SSIM evaluates structural similarity from the perspective of luminance, contrast, and structure. Higher PSNR and SSIM values indicate better visual fidelity and structural preservation.

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TABLE X
ASRs (%) OF DIFFERENT METHODS ON TRANSFORMER-BASED MODELS

Dataset	Model	FGSM	BIM	CW	PGD	MIFGSM	Jitter	AutoAttack	FogFool
UCM	ViT-B/16	82.18	95.33	61.14	96.23	96.46	82.18	98.15	98.27
	Swin-T	81.34	96.81	95.02	97.61	98.42	94.28	99.51	99.13
NWPU	ViT-B/16	95.10	97.59	81.82	98.62	98.31	94.38	99.30	100.00
	Swin-T	92.67	98.93	93.14	98.74	98.77	98.58	99.76	99.88
Average		87.82	97.17	82.78	97.80	97.99	92.36	99.18	**99.32**

TABLE XI
CNN-TO-TRANSFORMER TASRs (%) ON THE UCM AND NWPU DATASETS

Dataset	Model	MI-FGSM	DI-FGSM	TI-FGSM	NI-FGSM	SI-NI-FGSM	VMI-FGSM	VNI-FGSM	FogFool
UCM	ViT-B/16	4.95	9.65	14.36	4.95	5.94	9.16	8.91	29.21
	Swin-T	14.43	26.12	29.60	14.18	18.91	25.37	26.87	39.80
NWPU	ViT-B/16	16.03	25.72	36.00	13.40	15.31	24.76	25.00	69.98
	Swin-T	45.74	67.02	71.87	41.61	42.91	62.53	68.09	78.72
Average		20.29	32.13	37.96	18.54	20.77	30.46	32.22	**54.43**

TABLE XII
AVERAGE PSNR AND SSIM OF ADVERSARIAL EXAMPLES ON UCM AND NWPU DATASETS

Attack Method	UCM		NWPU	
	PSNR	SSIM	PSNR	SSIM
DI-FGSM	31.28	0.8364	31.35	0.8419
MI-FGSM	33.34	0.8785	33.46	0.8753
NI-FGSM	32.16	0.8321	32.23	0.8373
SI-NI-FGSM	31.14	0.8326	31.15	0.8384
TI-FGSM	33.26	0.8627	33.38	0.8606
VMI-FGSM	31.23	0.8362	31.29	0.8415
VNI-FGSM	31.13	0.8334	31.13	0.8368
FogFool	**35.34**	**0.9225**	**34.62**	**0.9201**

As reported in Table XII, FogFool achieves the highest PSNR and SSIM values on both datasets. On the UCM dataset, FogFool obtains 35.34 PSNR and 0.9225 SSIM, outperforming the strongest baseline MI-FGSM by 2.00 dB in PSNR and 0.0440 in SSIM. On the NWPU dataset, FogFool also achieves the best visual quality, with 34.62 PSNR and 0.9201 SSIM. These results indicate that the proposed fog-based perturbations better preserve the visual structure of RS images compared with conventional gradient-based perturbations.

The superior image quality can be attributed to the spatially coherent and mid-to-low-frequency characteristics of the optimized fog masks. Instead of introducing scattered pixelwise noise, FogFool embeds adversarial information into natural fog-like atmospheric patterns, which reduces abrupt local distortions and maintains higher structural consistency with the original images. Therefore, FogFool achieves a favorable balance between attack effectiveness and visual fidelity.

8) Ablation Study on Perlin Noise and Optimization: To clarify the practical contribution of Perlin noise and the gradient-guided optimization process, we conduct an ablation study with different fog-mask initialization strategies. In particular, we compare four types of initial fog masks, including uniform random noise, Gaussian random noise, single-scale Perlin noise, and FBM-Perlin noise. For each initialization strategy, we further evaluate its nonoptimized version and optimized version, where "Opt." denotes the same gradient-guided fog-mask optimization used in FogFool. Therefore, this ablation study examines whether Perlin noise only improves visual naturalness or also provides a more effective structured initialization for adversarial optimization.

For quantitative evaluation, we report the low-frequency energy ratio (LFER), ASR, PSNR, SSIM, and visual naturalness of each variant. LFER is computed as the proportion of perturbation energy located in the low-frequency region of the Fourier spectrum, which reflects the frequency distribution of the generated perturbations. ASR is averaged over all ten victim models on the UCM dataset.

As shown in Fig. 11, uniform and Gaussian noise tend to produce spatially scattered or unnatural fog patterns, whereas Perlin-based initialization generates more coherent and visually plausible atmospheric structures. In particular, FBM-Perlin fog exhibits multiscale fog-like patterns, which better resemble natural atmospheric effects in RS imagery.

The quantitative results in Table XIII further demonstrate the respective roles of initialization and optimization. Without adversarial optimization, single-Perlin fog and FBM-Perlin fog achieve very low ASRs of 2.82% and 3.08%, respectively, indicating that Perlin noise itself does not inherently carry strong adversarial attributes. Its main role is to provide a spatially coherent and visually natural initialization. After optimization, all variants obtain significantly higher ASRs,

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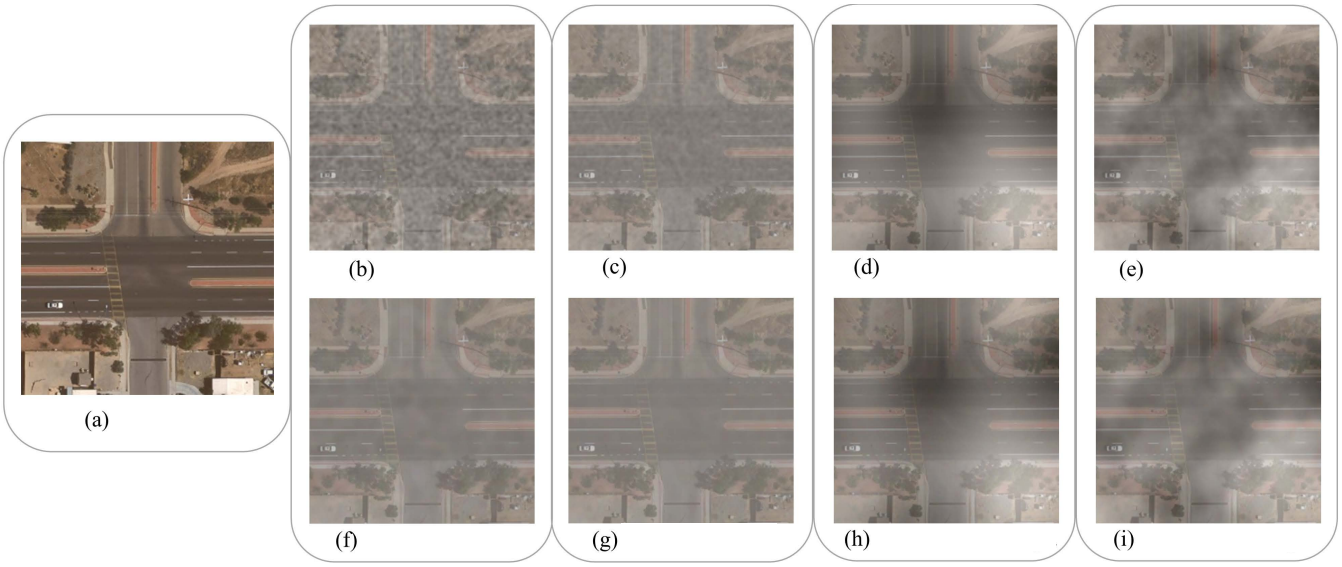


Fig. 11. Visual comparison of adversarial examples generated with different fog-mask initialization and optimization strategies. The leftmost image is the original clean image. The first row shows adversarial examples generated without adversarial optimization, including uniform fog, Gaussian fog, single-scale Perlin fog, and FBM-Perlin fog. The second row shows the corresponding optimized variants. Compared with uniform and Gaussian noise, Perlin-based fog masks exhibit more spatially coherent and natural fog-like structures, and the complete FBM-Perlin + optimization setting corresponds to the proposed FogFool. (a) Original. (b) Uniform Fog. (c) Gaussian Fog. (d) Single-Perlin Fog. (e) FBM-Perlin Fog. (f) Uniform + Opt. (g) Gaussian + Opt. (h) Single-Perlin + Opt. (i) FBM-Perlin + Opt.

TABLE XIII
ABLATION STUDY ON FOG-MASK INITIALIZATION AND OPTIMIZATION STRATEGIES

Variant	LFER	ASR	PSNR	SSIM	Naturalness
Uniform Fog	66.92%	17.27%	27.19	0.8419	Poor
Gaussian Fog	72.57%	11.04%	29.36	0.8568	Poor
Single-Perlin Fog	78.31%	2.82%	33.54	**0.9375**	Moderate
FBM-Perlin Fog	79.70%	3.08%	33.78	0.9338	Good
Uniform Fog + Opt.	81.48%	90.91%	29.38	0.8179	Poor
Gaussian Fog + Opt.	83.39%	91.17%	32.42	0.8427	Poor
Single-Perlin Fog + Opt.	86.27%	96.36%	**35.56**	0.9241	Moderate
FBM-Perlin Fog + Opt. (FogFool)	**86.67%**	**96.88%**	35.34	0.9225	Good

confirming that the adversarial capability mainly comes from the gradient-guided optimization process.

However, random noise cannot effectively replace Perlin-based initialization when both attack effectiveness and visual quality are considered. Although Uniform Fog + Opt. and Gaussian Fog + Opt. achieve ASRs above 90%, their PSNR, SSIM, and visual naturalness remain clearly inferior to the Perlin-based optimized variants. On the contrary, FBM-Perlin Fog + Opt., i.e., the complete FogFool, achieves the highest ASR of 96.88% and the highest LFER of 86.67% while maintaining high PSNR and SSIM values. These results indicate that FBM-Perlin initialization provides a more suitable structured search space for adversarial optimization, enabling FogFool to achieve a better balance between attack effectiveness, low-frequency dominance, and visual plausibility.

9) Defense Against Perlin Noise Fog: To further evaluate the robustness of the proposed FogFool under adversarial defense scenarios, we compare FogFool with two strong white-box baseline attacks, AutoAttack and PGD. First, adversarial examples are generated using the three attack methods to obtain the original ASRs. Subsequently, two commonly used defense preprocessing techniques—JC [58] and total variation minimization (TVM) [59]—are applied to mitigate adversarial perturbations. The ASRs are then re-evaluated after defense processing. For TVM, the reconstruction method is set to Bregman iteration [60], the pixel random drop rate is 0.5, and the TV regularization weight is 0.03. For JC, the image quality factor is set to 50. The experimental results are illustrated in Fig. 12.

As shown in Fig. 12, FogFool exhibits stronger resilience against both JC and TVM defenses compared with AutoAttack and PGD. Under JC, the ASRs of AutoAttack and PGD decrease substantially across most architectures, whereas FogFool maintains relatively higher ASR values. For instance, on ResNet50, the ASRs of AutoAttack and PGD decrease to 28.1% and 21.6%, respectively, while FogFool retains 30.1%. The performance gap further widens under TVM defense. On DenseNet121, the ASRs of AutoAttack and PGD decrease to 22.8% and 23.8%, respectively, whereas FogFool achieves

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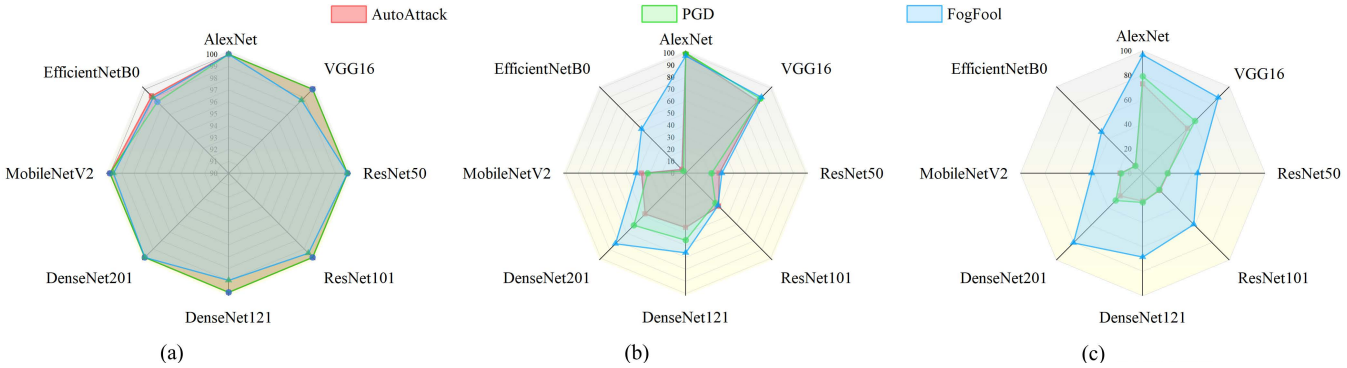


Fig. 12. Radar charts comparing the ASRs of AutoAttack, PGD, and the proposed FogFool method across eight models. (a) Raw ASR without defense. (b) ASR under JC defense. (c) ASR under TVM defense. Results are evaluated on the UCM dataset.

68.3%, indicating significantly stronger robustness against smoothing-based defenses.

This improved resilience can be attributed to the spatially coherent and low-frequency characteristics of the Perlin noise-based fog perturbations. Unlike pixelwise gradient perturbations, which often contain high-frequency components that are easily suppressed by compression or denoising operations, FogFool resembles natural atmospheric scattering patterns. Such structured perturbations are less sensitive to frequency-domain filtering and smoothing processes, thereby maintaining higher attack effectiveness after defense preprocessing. These results demonstrate that FogFool not only achieves competitive attack performance but also preserves stronger robustness under common adversarial defense strategies.

10) Why Perlin Noise Fog Works: To investigate the underlying mechanism of FogFool, we employ Grad-class activation map (CAM) [61] to visualize the CAMs of ResNet50 before and after applying Perlin noise fog perturbations. Grad-CAM highlights the discriminative regions that contribute most significantly to the model's prediction.

As shown in Fig. 13, the introduction of Perlin noise fog leads to a substantial redistribution of attention in the feature space. In the original images, the network focuses primarily on semantically meaningful objects, such as the airplane in the leftmost example. However, after adding structured fog perturbations, the high-activation regions shift toward surrounding background areas, while the object-related activations are weakened.

Importantly, the global structural integrity of the objects remains largely preserved in the adversarial examples, indicating that the perturbation does not rely on visually destructive distortions. Instead, the Perlin noise fog alters intermediate feature representations and attention allocation, thereby disrupting the discriminative cues used by the classifier. This structured redistribution of activation regions explains why the proposed fog-based perturbations can effectively mislead the neural network while maintaining perceptual plausibility.

11) Scope and Limitations: FogFool is designed and evaluated as a digital-domain adversarial perturbation framework. Although its fog-mask construction is inspired by atmospheric phenomena and produces spatially coherent fog-like patterns, the current method optimizes the full image

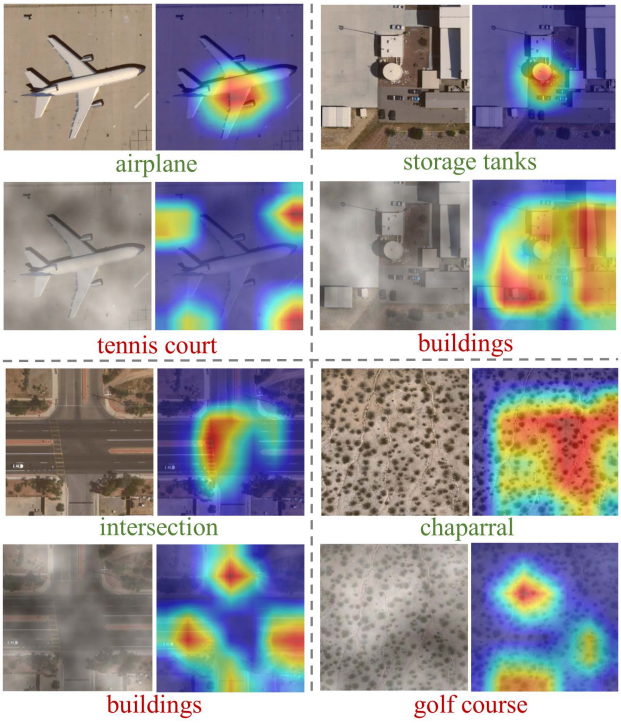


Fig. 13. Grad-CAM visualizations of ResNet50 on the UCM dataset. From (top to bottom), the rows show original images with their CAMs, and adversarial examples with their CAMs. The green labels indicate the ground-truth categories, while the red labels indicate the predictions of the target models.

after acquisition and does not guarantee direct physical-world realizability. In particular, variations in scene depth, viewpoint, illumination, atmospheric dynamics, and sensor response are not explicitly modeled. Future work will investigate depth-aware atmospheric scattering, environment-aware optimization, expectation-over-transformation strategies, and physical image-capture experiments to evaluate the feasibility of fog-based adversarial perturbations under real sensing conditions.

V. Conclusion

In this article, we proposed FogFool, a physically plausible adversarial attack framework for RS image classification. Unlike conventional pixelwise perturbation methods constrained by ℓ_p-norms, FogFool restricts the perturbation space

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to structured atmospheric fog patterns generated via multioctave Perlin noise and FBM. By integrating procedural fog simulation with gradient-guided optimization, FogFool is capable of producing both untargeted and targeted adversarial examples while maintaining strong spatial coherence and visual realism. Extensive experiments conducted on two benchmark RS datasets demonstrate that FogFool achieves competitive or superior ASRs compared with state-of-the-art white-box and transfer-based attack methods. FogFool maintains consistently high ASRs across diverse network architectures, including lightweight and deep convolutional models. Furthermore, parameter analyses confirm that the fog blending coefficient and optimization iterations can effectively balance visual plausibility and attack effectiveness. The generated adversarial examples exhibit natural atmospheric characteristics, avoiding the artificial high-frequency artifacts commonly observed in traditional attacks.

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